be the possibility of a far-distant, instantaneous influence affecting your results. Einstein, in particular, emphasized that locality must be a core principle of science, not given up unless absolutely necessary, for exactly this reason. “Without such an assumption of the mutually independent existence (the ‘being-thus’) of spatially-distant things, an assumption which originates in everyday thought, physical thought in the sense familiar to us would not be possible,” he wrote. “Nor does one see how physical laws could be formulated and tested without such a clean separation.… The complete suspension of this basic principle would make impossible the idea of the existence of (quasi-) closed systems and, thereby, the establishment of empirically testable laws in the sense familiar to us.”

Even putting Einstein’s philosophical concerns aside, Einstein’s scientific work made it clear that locality was a key feature of the world. According to Einstein’s special relativity, physical objects can’t be pushed up to or past the speed of light, on pain of a whole host of paradoxes involving infinite energy. You might try to get around this by finding something that is already moving faster than light—but no such object has ever been found. Indeed, relativistic particle physics states that such objects would be spectacularly unstable, precluded from existence by their own special infinite-energy paradoxes. And even if you somehow get around these problems and manage to send a signal faster than light, you still run the risk of paradox. Relativity dictates that merely sending a faster-than-light signal would immediately make it possible to construct a “tachyonic antitelephone” that would let you send messages back in time.

But Bell’s theorem doesn’t mean we call ourselves yesterday or send a DeLorean to 1955. Bell and others later proved that it was impossible to use quantum entanglement for faster-than-light signaling. And the specific sort of nonlocality displayed by entangled particles is so delicate and subtle, appearing only under such specific conditions, that it doesn’t pose the sort of existential threat to science itself that Einstein had feared. But the fact remains that in a world where special relativity has easily passed every single test we’ve thrown at it—a world that appears local—the specter of nonlocality raised by Bell’s theorem is profoundly disturbing. If the quantum prediction for Bell’s experiment is correct, and Bell’s inequality is violated, then something is nonlocal, and locality is merely an illusion. That suggests a need for a radical revision of our conception of space and time,